

DATA GOVERNANCE MATURITY & ROI EXECUTIVE REPORT

Platform: **ALATION** | Scope: Enterprise Implementation | Evaluation Date: 2026-06-06

Executive Summary: An automated audit-ready maturity assessment was conducted on **50** data assets managed by Alation. The implementation achieves a **Maturity Score of 2.50/5.0** (Documentation Health: 73.4%, DQ Pass Rate: 72.8%). Financially, the program has generated **\$24,066,816.46** in realized business value against an estimated annual operating cost of **\$120,000.00**, yielding a net realized value of **\$23,946,816.46**. Additionally, the engine has identified **\$12,688,896.48** in unrealized, actionable opportunity savings that can be unlocked through targeted remediation efforts.

1. Discipline-Level Data Governance Maturity Dashboard

Scores are computed across critical governance disciplines mapped directly from catalog metadata using transparent, auditable rules. Health indicators are nested under their respective disciplines:

Governance Discipline / Health Indicator	Weight	Raw Value / Score	Status
Metadata Management	50%	2.60 / 5.0	Amber
■ Documentation Coverage	30%	10.0% (Score: 1.0)	Red
■ Ownership Coverage	30%	72.0% (Score: 3.0)	Amber
■ Glossary Linkage	20%	54.0% (Score: 2.0)	Red
■ Classification Coverage	20%	98.0% (Score: 5.0)	Green
Data Quality	50%	2.40 / 5.0	Action Needed
■ DQ Rule Coverage	40%	68.0% (Score: 3.0)	Amber
■ DQ Pass Rate	60%	72.8% (Score: 2.0)	Red
Overall Maturity Index	100%	2.50 / 5.0	Amber

2. Program Financial Performance & ROI Analysis

The value realized through metadata-driven automation and risk mitigation compared to software licensing and overhead operating costs:

Financial Metric Category	Amount (\$)	Value Explanation
Annual Operating Cost	\$120,000.00	Fixed software licensing and support overhead cost.
Realized Discovery Savings	\$23,039,383.50	Productivity savings from faster data discovery.
Realized DQ Incident Avoidance	\$810,000.00	Savings from avoiding pipeline failures and business downtime.
Realized Risk Avoidance	\$179,400.00	Mitigated risk of security breaches and regulatory non-compliance.
Realized Compute Optimization	\$38,032.96	Warehouse credit waste avoided via data quality monitoring.

Net Realized Program Value	\$23,946,816.46	Net financial return of the program.
Return on Investment (ROI)	19955.68%	Program efficiency ratio.
Unrealized Opportunity Value	\$12,688,896.48	Value pipeline achievable through remediation tasks.

3. Executive Action Plan & Remediation Pipeline

The engine has identified key prioritized actions to reduce costs and secure sensitive corporate data:

Category	Asset Name	Risk / Telemetry Details	Recommended Action	Value Impact
Storage / ROT	executive_kpi_125	Size: 136.0 GB (579d stale)	Decommission or archive storage	Saves \$6.53/yr
Storage / ROT	user_retention_129	Size: 219.2 GB (374d stale)	Decommission or archive storage	Saves \$2.60/yr
Storage / ROT	temp_customer_export_2024	Size: 120.0 GB (191d stale)	Decommission or archive storage	Saves \$1.43/yr
Compliance / PII	temp_customer_export_2024	Risk: 80.0/100 Queries: 3/mo	Assign Steward & classification tags	Mitigates \$120,000.00 risk
Compliance / PII	marketing_attribution_123	Risk: 80.0/100 Queries: 8/mo	Assign Steward & classification tags	Mitigates \$120,000.00 risk
Compliance / PII	backup_restore.tar.gz_109	Risk: 75.0/100 Queries: 1833/mo	Assign Steward & classification tags	Mitigates \$112,500.00 risk
Trust / Low DQ	marketing_attribution_123	DQ Pass Rate: 0.0% Queries: 8/mo	Implement validation rules in pipeline	Avoids debug costs
Trust / Low DQ	temp_customer_export_2024	DQ Pass Rate: 0.0% Queries: 3/mo	Implement validation rules in pipeline	Avoids debug costs
Trust / Low DQ	orders_138	DQ Pass Rate: 0.0% Queries: 1528/mo	Implement validation rules in pipeline	Avoids debug costs

4. Maturity Assessment Strengths & Gaps

Key Governance Strengths (Green):

- 98.0% classification coverage (Status: Green, Target: >= 90%)

Identified Governance Gaps (Amber/Red):

- 10.0% documentation coverage (Status: Red, Target: >= 90%)
- 72.0% ownership coverage (Status: Amber, Target: >= 90%)
- 54.0% glossary linkage (Status: Red, Target: >= 90%)
- 68.0% dq rule coverage (Status: Amber, Target: >= 85%)
- 72.8% dq pass rate (Status: Red, Target: >= 95%)

5. Prioritized Action Plan & Recommendations

Actionable remediation roadmap to accelerate maturity and unlock business value:

Recommendation 1: Increase business description coverage.

Rationale: Over 50% of assets lack standard descriptions, making self-service data discovery slow and redundant.

Business Impact: Improves data discovery productivity and reduces onboarding overhead.

Maturity Improvement: Increases Metadata Management maturity from 2.50 to 2.90.

Recommendation 2: Link data catalog assets to the standardized Business Glossary.

Rationale: Without glossary mapping, columns have inconsistent meaning across domains.

Business Impact: Aligns cross-team terminology and improves report accuracy.

Maturity Improvement: Increases Metadata Management maturity from 2.50 to 2.90.

Recommendation 3: Establish data quality rules for Critical Data Elements.

Rationale: Popular datasets are run without data quality validation rules configured.

Business Impact: Avoids pipeline failures and business decision-making errors.

Maturity Improvement: Increases Data Quality maturity from 2.50 to 2.90.

APPENDIX: ROI METHODOLOGY & ASSUMPTIONS

To ensure auditability, the calculations in this executive assessment are based on the following models and parameter mappings. These baselines are grounded in industry studies conducted by Gartner, Forrester, IBM Security, and standard cloud platform metrics.

Model Baseline Constants & Citation Sources

Parameter Constants	Value	Research Context / Citation Source
Loaded Analyst Rate	\$75.00/hr	U.S. Labor Bureau senior loaded rate averages (1.35x multiplier).
Hours Saved per Search	3.5 hrs	Forrester TEI data catalog productivity metrics.
Annual Storage Cost / GB	\$0.24	Standard hot cloud object tier blended averages (AWS, Azure).
Data Quality Error Cost	\$15,000.00	Gartner & Monte Carlo operational cost studies.
Compliance Breach Penalty	\$150,000.00	IBM Cost of Data Breach localized/tier-1 infraction fine.

Equations & Derivations

- Productivity Savings:** (Annual Queries * 10% search ratio) * 3.5 hrs saved * Analyst loaded rate * (Doc Score / 100). Grounded in Forrester's TEI frameworks.
- ROT Decommissioning:** Identified as size > 0, queries < 5/mo, and last accessed > 180 days. Savings = GB Size * \$0.24/yr. Grounded in AWS/Azure storage tiers.
- DQ Incident Avoidance:** (Unmonitored baseline [4.0] - Current incidents) * \$15k cost. Active DQ profiling drops current errors to 2.0 (for DQ >= 80%) or 0.0 (for DQ >= 95%). Grounded in Gartner data quality impact surveys.
- Risk Avoidance:** (Baseline breach prob. [5.0%] - Current prob.) * \$150k breach penalty. Controls (ownership and classification) reduce probability to 1.0% (0.2% with active DQ checks). Grounded in IBM Security breach statistics and DAMA DMBOK principles.
- Compute Optimization:** Snowflake credit waste avoided on bad/unoptimized queries, calculated based on the active warehouse sizes, average credit rates (\$3.00/credit), and active DQ monitoring rules.
- Net Program Value:** Net Value = Total Realized Savings - Annual Operating Cost.
- Return on Investment (ROI) %:** ROI % = (Net Realized Program Value / Annual Operating Cost) * 100.
- Unrealized Opportunity Value:** Sum of remaining discovery opportunity + DQ opportunity + Risk mitigation + ROT storage decommissioning savings.

Discipline Maturity & Indicator Assessment Formulas

- Documentation Coverage %:** (Assets with description >= 50 characters / Total Assets) * 100. Thresholds: [40, 60, 75, 90]%.
- Ownership Assignment %:** (Assets with >= 1 owner / Total Assets) * 100. Thresholds: [40, 60, 75, 90]%.
- Glossary Linkage %:** (Assets with >= 1 glossary term / Total Assets) * 100. Thresholds: [40, 60, 75, 90]%.
- Classification Coverage %:** (Assets with >= 1 classification / Total Assets) * 100. Thresholds: [40, 60, 75, 90]%.
- DQ Rule Coverage %:** (Assets with >= 1 DQ rules run / Total Assets) * 100. Thresholds: [25, 50, 70, 85]%.
- DQ Pass Rate %:** (Total rules passed / Total rules run) * 100. Thresholds: [70, 80, 90, 95]%.
- Maturity Scoring Step-mapping:** Raw percentages map to 1-5 levels based on the thresholds: Score = 1.0 (if < thresholds[0]), 2.0 (if < thresholds[1]), 3.0 (if < thresholds[2]), 4.0 (if < thresholds[3]), 5.0 (if >= thresholds[3]).
- Metadata Management Maturity:** Weighted average of its indicators: Documentation (30%), Ownership (30%), Glossary Linkage (20%), and Classification (20%).

- **Data Quality Maturity:** Weighted average of its indicators: Rule Coverage (40%) and Pass Rate (60%).
- **Overall Maturity Index:** Weighted rollup of Metadata Management (50%) and Data Quality (50%).